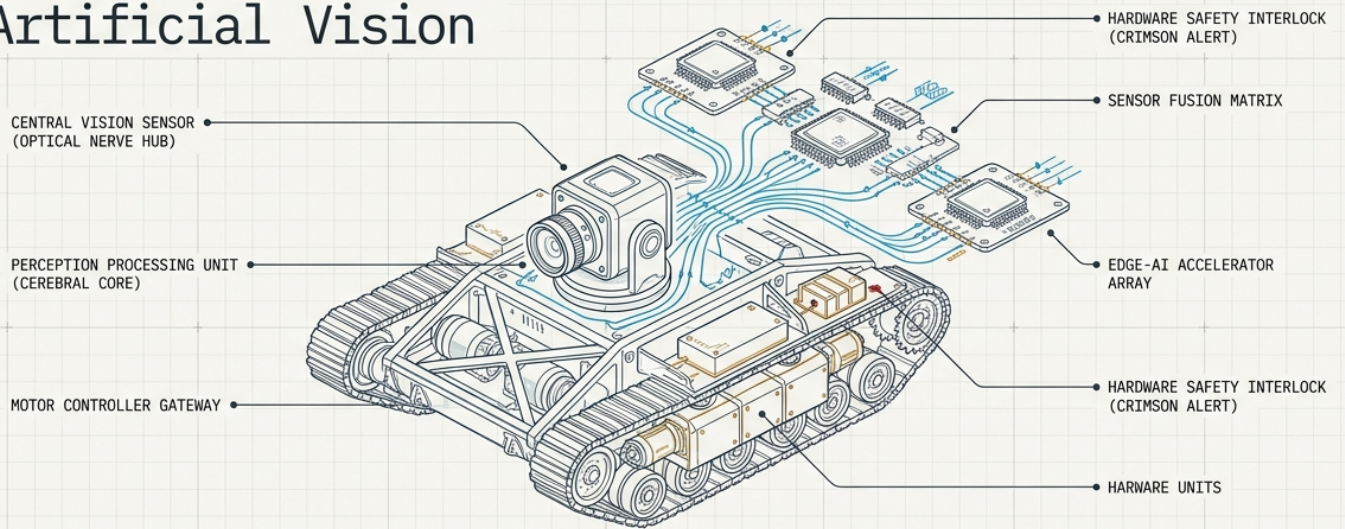


Building a Synthetic Organism for Artificial Vision

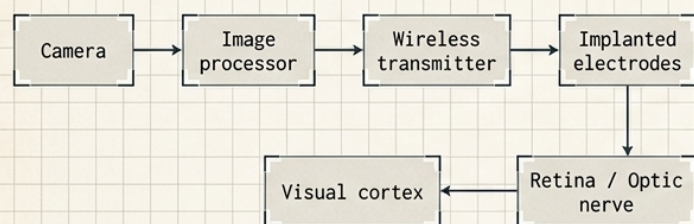


TANKOS VER. 3.2 // EDGE-AI NETWORK // MULTI-LLM COGNITION // DETERMINISTIC SAFETY // SCALABLE FOUNDATION

The Tank Project is a from-scratch autonomous robotic operating system (TankOS) that distributes intelligence across an edge-AI network. By merging deterministic hardware safety with multi-LLM cognition, it creates a scalable foundation for both autonomous robotics and wearable bionic assistants.

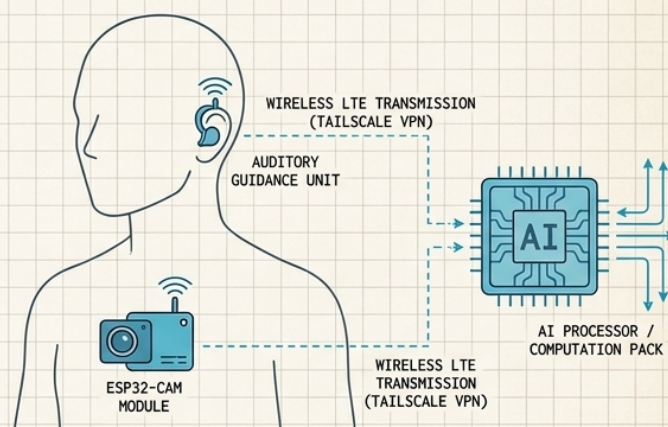
The medical mechanics of restoring visual perception

The Biological Baseline



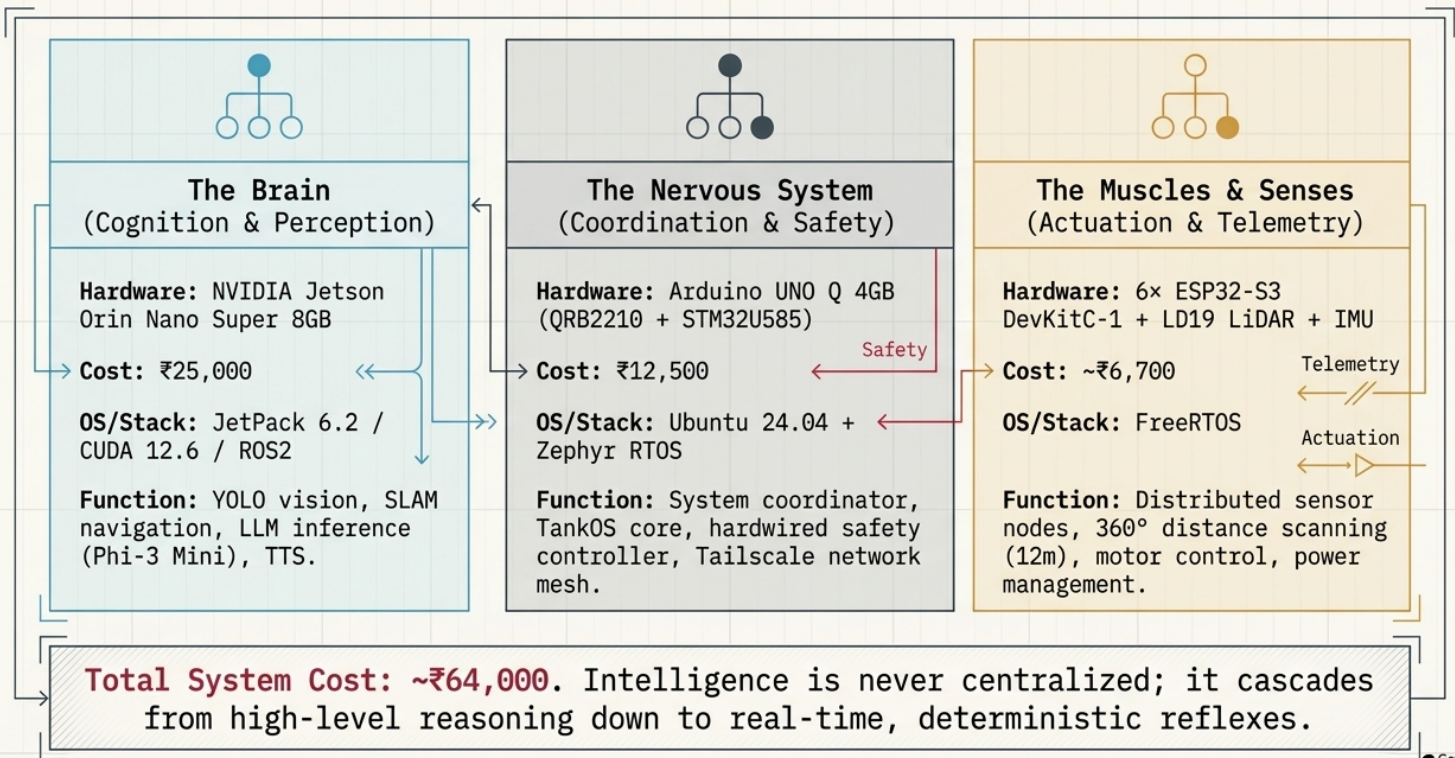
Artificial vision does not restore normal vision. It converts visual information into electrical impulses, bypassing damaged photoreceptors to provide light perception, object localization, and improved mobility.

The Cybernetic Approach

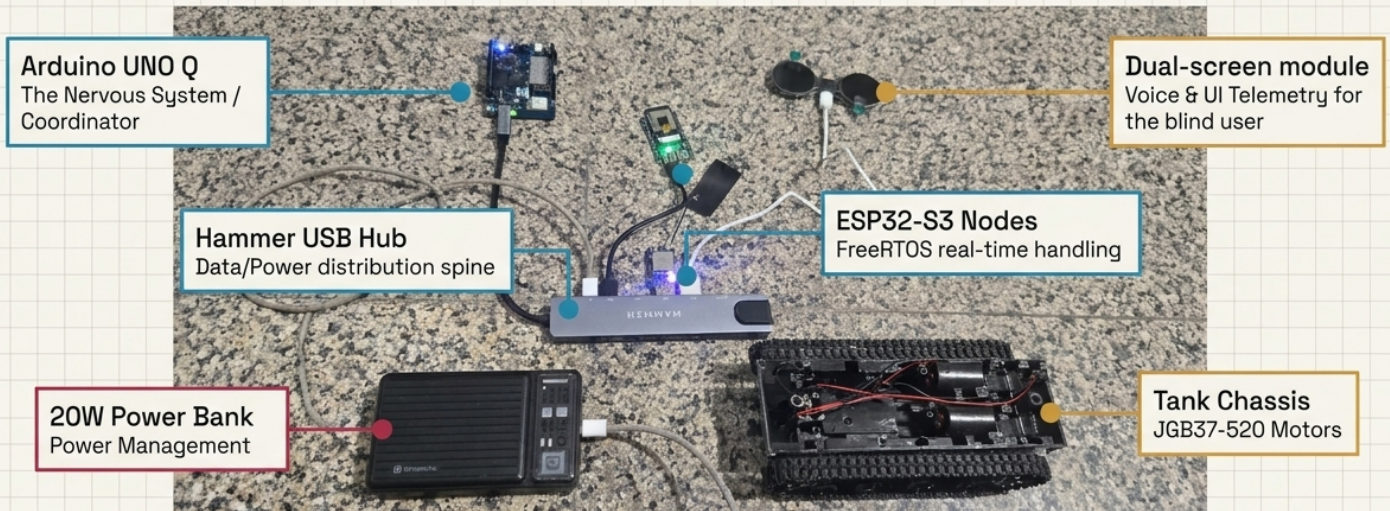


Dr. Shashi Gupta's implementation utilizes an ESP32-CAM module to capture the environment. It bypasses biological implants by transmitting data via a USB LTE modem (Tailscale VPN) to a centralized AI processor, translating visual data into auditory spatial guidance and automated locomotion.

Distributing intelligence across a three-layer biological architecture

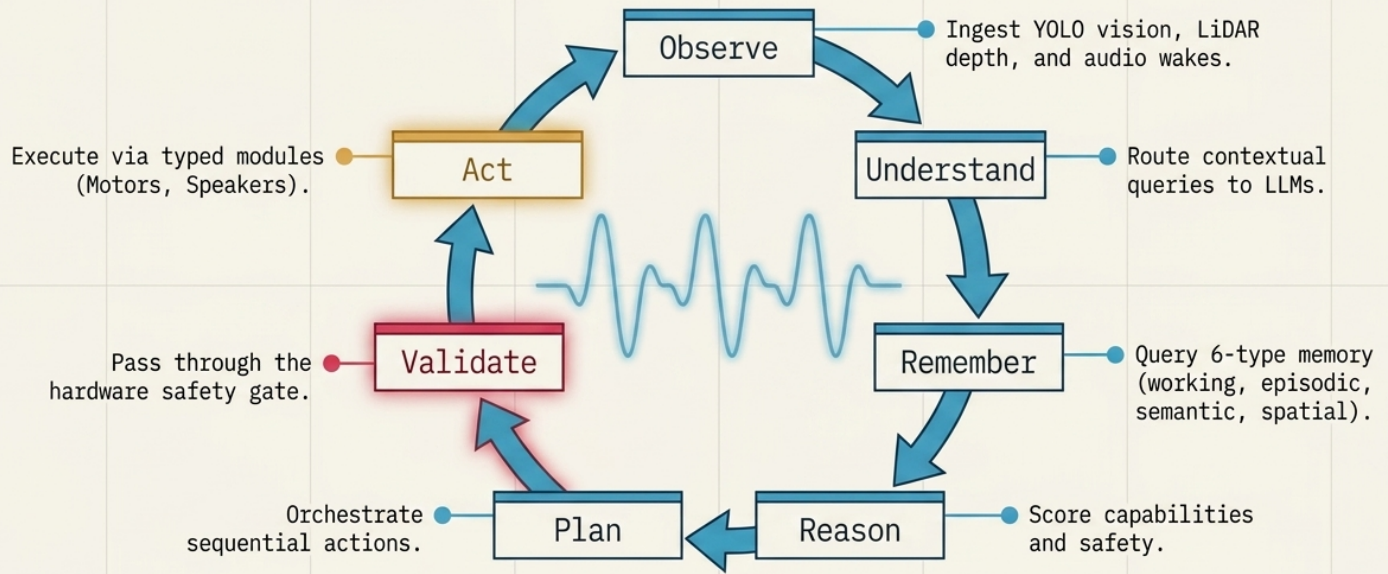


The physical blueprint of the distributed edge-network



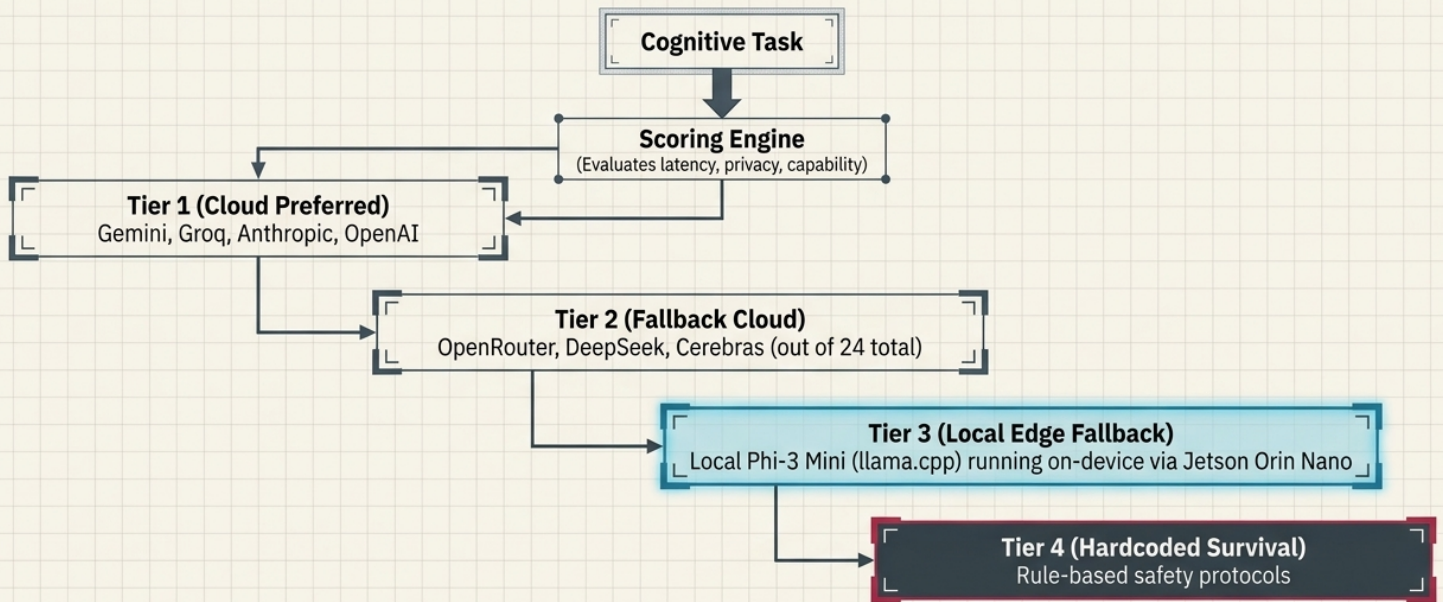
A modular hardware topology allows the 'Brain' (Jetson) to exist remotely via LTE, while the lightweight 'Nervous System' (UNO Q) and 'Muscles' (ESP32) remain on-site with the user or the robotic chassis.

The cognitive metabolism of TankOS



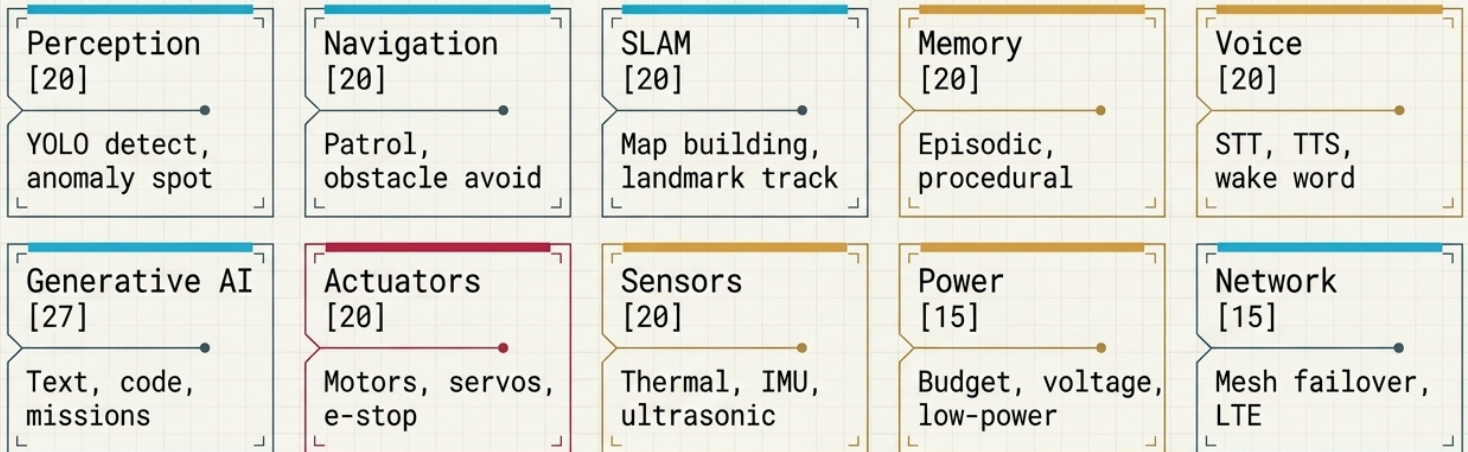
Unlike traditional linear scripts, TankOS operates continuously. Every sensory input passes through a mandatory seven-stage cognitive loop before any physical actuation occurs, ensuring the robot constantly reassesses its environment.

Multi-provider synaptic routing ensures zero downtime



Intelligence cannot rely on a single failure point. TankOS dynamically routes tasks across 24 distinct LLM providers. If network connectivity drops (e.g., in a remote environment), the system instantly falls back to local on-device inference via Phi-3, ensuring the visually impaired user is never left stranded.

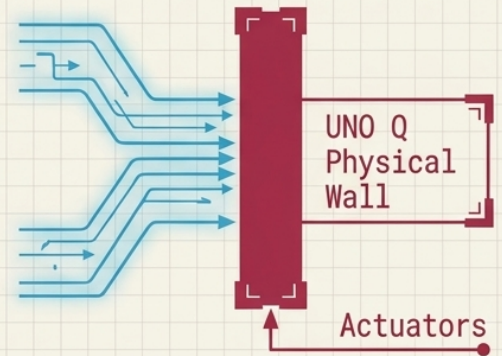
A 327-module digital skill tree governs all behavior



Every function is a typed, permissioned, LLM-callable module. The AI does not write code to move; it calls precise, pre-validated nodes from an 18-group library, managed entirely through the 500-feature Android TV GUI.

The Safety Immune System restricts AI autonomy by risk tier

Risk Level	Example Action	Required Gate
R0 (Docs/Comments)	Read vision capture	Auto-approval
R1 & R2 (UI/Performance)	UI layout updates	Automated tests & reversibility checks
R3 (AI Behavior)	Prompt adjustments	Sandbox simulation + Review
R4 (Robot Navigation)	Go-to waypoint	Hardware-in-the-loop + Human confirmation
R5 (Critical Operations)	Emergency Stop / Motor Limits	Hardwired (AI cannot override)



The auto-evolution engine allows TankOS to improve its own code, but intelligence is subordinate to physics. A physical "Wall" resides on the UNO Q MCU—if an AI plan triggers an R5 threshold, a hard-stop intercepts the signal before it reaches the ESP32 muscles.

Translating biological prosthetics into cybernetic architecture

Biological Prosthetic Components

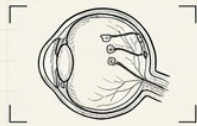


Camera mounted on spectacles

External Image Processor

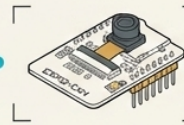


Wireless Transmitter Coil

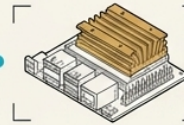


Implanted Electrodes / Retinal Stimulation

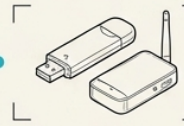
Cybernetic mapping diagram



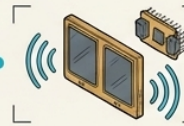
ESP32-CAM module worn by the user



Jetson Orin Nano (YOLOv8n processing depth & objects)



USB LTE Modem routing via Tailscale VPN



Dual-screen speaker module (Audio spatial feedback) + ESP32 Actuators (haptic steering)

By externalizing the 'Visual Cortex' to the Jetson Orin Nano and replacing surgical electrodes with auditory and haptic feedback nodes, the Tank architecture effectively acts as a non-invasive, high-resolution cortical prosthesis.

The Bionic Assistant: Wearable sensory telemetry

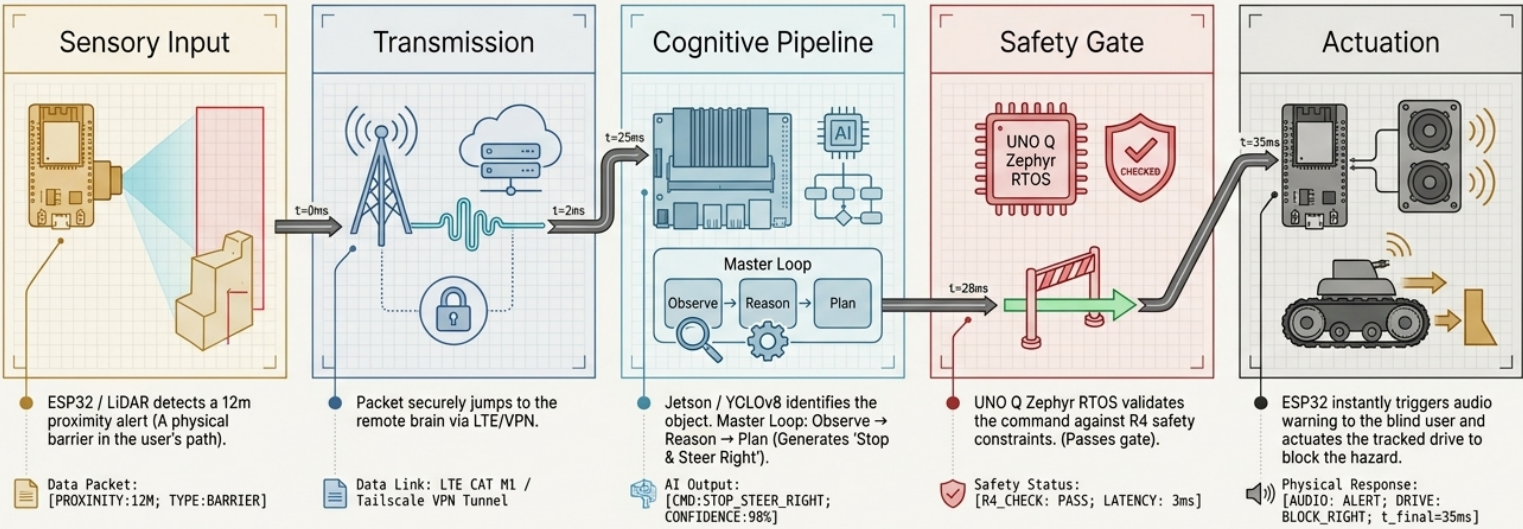


ESP32-S3
Node

Power
Bank

As described by Dr. Gupta, the end-user deployment unchains the intelligence from the tank chassis. The visually impaired user carries only a lightweight ESP32-S3 camera node and a standard power bank. The heavy computational lifting (LLM reasoning, SLAM) is handled remotely, ensuring the wearable device remains unobtrusive, cool, and highly energy-efficient.

Synthesis: Milliseconds from perception to preservation



A single byte of data traverses the entire biomimetic network. It is captured by the digital senses, analyzed by multi-LLM cognition, aggressively vetted by a hardwired safety cortex, and executed as physical protection—proving the architecture's speed, safety, and real-world utility.